Temi Adejare

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Understanding Diabetes Risk Using Health and Lifestyle Indicators

I will use the CDC Health Indicators Dataset, which is also available on Kaggle. This dataset includes health and lifestyle factors such as BMI, physical activity, and smoking status. I chose this dataset because I believe it takes into account many of the reasons why people may develop diabetes. Although the disease has become more treatable and better understood due to increased funding, this is not the case in all regions, and it still remains a real-world public health problem. Additionally, this dataset provides a large sample size, which makes it well-suited for classification analysis.

Methods:

- Logistic Regression because some of the features in the dataset (such as high blood pressure and high cholesterol) are binary. This model will look at how each feature is related to the likelihood of having diabetes. It is also easy to interpret, which makes it useful for understanding how different factors influence prediction.
- Decision Tree & Random Forest: This will be used for more complex patterns I notice in the data. For instance, it can help predict situations that may potentially increase diabetes risk, such as having a high BMI and low physical activity. I will use random forest because it uses multiple trees and combines their predictions.

- K-NN: This model will be used to help look at the people with similar characteristics in this dataset and use their outcomes to predict whether a new person has diabetes.

Research Question: Which machine learning model performs best at predicting diabetes risk, and how do lifestyle/health factors compare to socioeconomic factors in their impact on model performance?

Hypothesis

- Random forest will perform the best among the selected models because of its ability to combine multiple decision tress. It will perform better at capturing complex relationships within the dataset
- Lifestyle and health-related features will likely have a greater impact on prediction performance than socioeconomic features. This is because factors such as BMI and physical activity are more directly observable and are commonly associated with diabetes risk. However, I do believe that including both lifestyle/health and socioeconomic features together will produce the best model performance, since socioeconomic factors can still affect things like access to healthcare.

What I hope to learn: I have an interest in healthcare technology, and it is a field I hope to pursue professionally, so I hope to gain a better understanding of how different machine learning models perform on a real-world healthcare dataset. I want to learn how to compare models and evaluate their performance so that when I work on future projects, it will be easier for me to decide which model to use. I also hope to get a better understanding of how different types of features, such as lifestyle/health factors and socioeconomic factors, influence model predictions.

Finally, I want to improve my overall machine learning skills so that I can be prepared for full-time roles and continue to advance in my education.

Works Cited

CDC Diabetes Health Indicators Dataset — UCI Machine Learning Repository (ID 891) https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators